Network size and synapses against the group requirements

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I. Aim

The synapse study (2026-10-17) chose a network size and synapse format by energy and validation F1 on our own test rules. This report re-measures the best of those designs against the *group* requirements (Table II) on the *same* test sets as the other designs in the report, so that all designs can be compared directly.

II. System and network

Fig. 1 shows the hardware. The PC turns each 1 s window of audio into a 24×32 log-mel spectrogram every 250 ms. The ARM only relays it; the PicoRV32 soft core in the FPGA fabric runs the network from block RAM. Three fabrics were measured: plain RV32IM, RV32IM with the dense `kdot` coprocessor, and `kdot` with the sparse `ksdot` extension that skips pruned synapses.

Fig. 2 shows the network. Each window is one input vector of 768 values. H leaky integrate-and-fire neurons receive it through the input synapses W_1, integrate for 12 steps and spike; a linear readout of their spike counts gives a "Sheila" score. A word is detected when the score passes a threshold in 2 of 3 consecutive windows. The input synapses are 99.5 % of the memory and almost all of the work, so the designs differ in four ways: the number of neurons H, the bits per synapse, the number of synapses each neuron keeps (K, by pruning), and the hardware that executes them.

III. Training

Fig. 3 shows how every design was trained. Batches hold equal numbers of "Sheila" and other clips, 30 % of them synthetic near misses ("sh", "shee", "shee"+word, …) cut from real recordings. The 35 epochs end with 10 epochs of quantisation-aware training, so the integer network on the board loses nothing; pruned designs remove their smallest synapses gradually in epochs 4–20.

L1/L2 regularisation. No L1 or L2 penalty term was added to the loss (Table I). The optimiser's weight decay is the decoupled form of L2, and a small spike-count penalty acts like L1 on the neuron activity. An explicit L1 term was not used because sparsity is enforced directly: top-K pruning gives every neuron exactly K synapses, a regular pattern the `ksdot` hardware can exploit, whereas L1 gives irregular zeros. Overfitting is not the limit either: quality is flat from 16 neurons on, so the networks are limited by capacity, not by memorising the data.

TABLE I
Regularisation used. No explicit L1 or L2 term is added to the loss.

Technique	Setting	Acts on	Comparable to
Weight decay (AdamW)	10^{-3}	all weights	L2, decoupled
Spike penalty	$10^{-4}\times$ spikes	activity	L1 on activity
Top-K pruning	K per neuron	input synapses	hard L0
QAT	last 10 epochs	weights	weight noise
Augmentation	30 % of batch	data	—
Gradient clipping	norm 5	updates	—

TABLE II
Requirements and how they are measured. Every test set is the one used for the other designs in this report (Mateo's evaluation harness).

Requirement	Measured on	Target
Live "Sheila" detected	212 test clips in noise, 250 ms hop, 2 of 3 windows	$\geq 85\,\%$
Other words accepted	3,000 test words, same set-up	$\leq 0.2\,\%$
False accepts per hour	19.0 h: all other test words + LibriSpeech test-clean/-other	≤ 2
Word end to decision	median on a 1 h test stream, plus compute and link	≤ 500 ms
Compute per request	worst case on the board, 100 MHz	≤ 200 ms
Model memory	firmware model section	≤ 144 KiB
Average power, listening	PL, Vivado power × board cycles	≤ 1 W

IV. Requirements and evaluation data

Training uses the official Speech Commands v2 training split (84,843 clips, 1,606 "Sheila"). Every requirement is measured with Mateo's evaluation harness, copied unchanged, on the test split (Table II). The stream threshold is fixed on *validation* data with his rule: the highest live detection with at most 0.2 % other words and 2 false accepts per hour; the test sets are then used once. Two checks confirm the set-up: his live test reproduces our earlier live numbers exactly for all 39 models at their old thresholds, and retraining a design reproduces its weights bit for bit. Clip F1 is reported on the full test split and on Damien's 3,212-clip subset used in the Encoding section. Compute, memory and energy come from the bit-exact board runs of the synapse study.

V. Results

Table III lists all 13 designs; Table IV compares the two chosen ones with Mateo's designs.

Detection. No design reaches 85 % live detection (Fig. 4). The best mean is $42.9 \pm 6.9\,\%$ (48 neurons, 8-bit) and no single model exceeds 50 %.

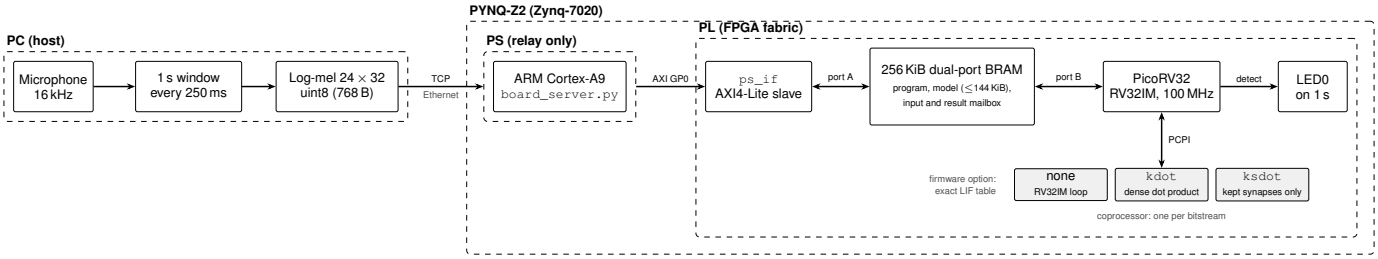


Fig. 1. System architecture. The PC records the audio and computes one 768-byte spectrogram every 250 ms; the ARM processing system (PS) only relays it. In the programmable logic (PL), the PicoRV32 core runs the spiking network from block RAM and lights LED0 on a detection. Three fabrics were measured: plain RV32IM, RV32IM with the dense `kdot` coprocessor, and with its sparse `ksdot` extension. Energy and the 1 W power requirement refer to the PL.

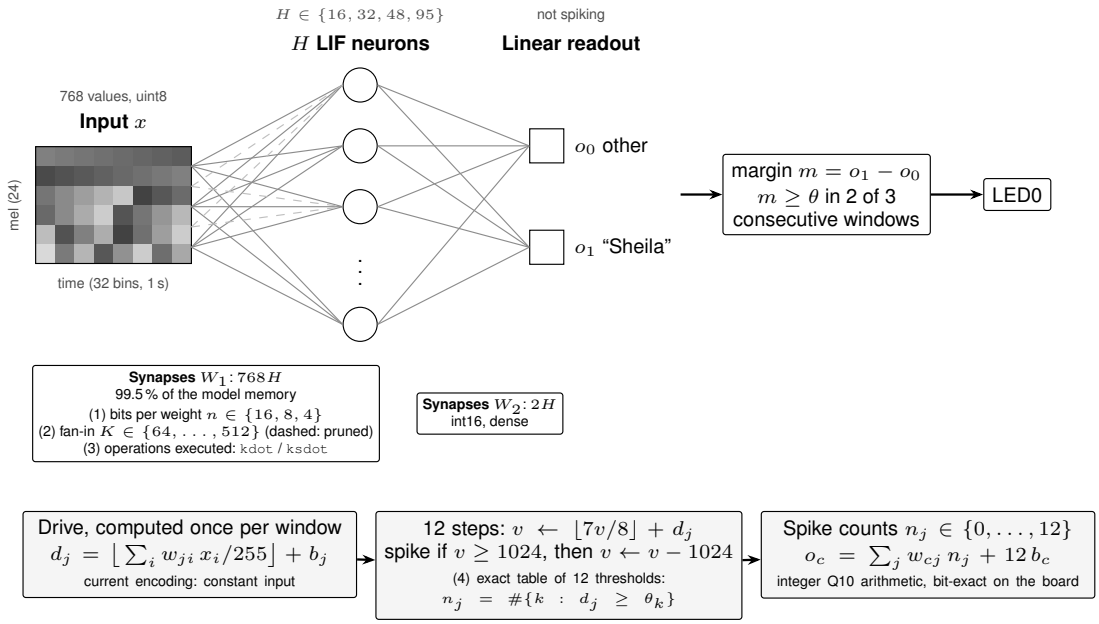


Fig. 2. The network: 768 spectrogram inputs, H leaky integrate-and-fire (LIF) neurons and a linear readout of their spike counts. Almost all cost is in the input synapses W_1; the study varies how many bits each synapse has (1), how many synapses each neuron keeps (2), and whether the hardware executes only the kept ones (3). Because the drive is constant, the 12-step neuron loop can be replaced by an exact table of 12 thresholds (4).

- *Other words:* every design accepts at most 0.07 %, well inside 0.2 %.
- *False accepts per hour:* the threshold keeps them at $\leq$2 on validation, but the test gives 1.6–4.1 (mean 2.8); only 2 of 39 models stay within 2. 86 % of the false accepts come from LibriSpeech running speech, which the networks never saw in training.
- *Price of the false-accept limit:* at the earlier S-study threshold the same models detected 49–67 %, but made 8–95 false accepts per hour. The $\leq$2 per hour limit, not the 0.2 % limit, sets the threshold.
- *No threshold reaches 85 %:* even at ten times the other-word limit, detection stays below 80 % (Fig. 8).

Clip F1 is unchanged by the new thresholds: 0.76–0.79 on the full test split and 0.81–0.85 on Damien's subset, where fewer negative clips raise the precision.

Size and synapses. From 32 neurons on, detection does not change with the number of neurons, the bits per synapse or the synapses kept per neuron by more than the seed spread (Fig. 5); only 16 neurons is clearly lower (26 %). The levers change the cost, not the detection.

Cost. Every design meets the five cost and latency requirements (Fig. 6, Fig. 7).

- Compute: at most 89 ms (95 neurons on plain RV32IM) against 200 ms; at most 3.9 ms with `kdot`.
- Memory: at most 143 KiB; a pruned network with 64 synapses per neuron needs 12.4 KiB.
- Word end to decision: 134–258 ms median against 500 ms.
- Power: 0.36–0.37 W for the PL as built (busy-polling), 0.09–0.19 W with clock gating, against 1 W. The 1 W budget allows 250 mJ per 250 ms window; one inference needs 45 μJ–25 mJ.

Choice: dense or pruned. The rule fixed beforehand picks the best dense and the best pruned design and keeps the pruned one unless the dense one is better by more than its seed spread (Fig. 9).

- Best dense: 48 neurons, 8-bit, `kdot`+table (the deployed model): 42.9 ± 6.9 %, 165 μJ, 72.4 KiB.
- Best pruned: 48 neurons, 64 synapses each, `ksdot`+table: 36.3 ± 6.1 %, 96.5 μJ, 12.4 KiB.

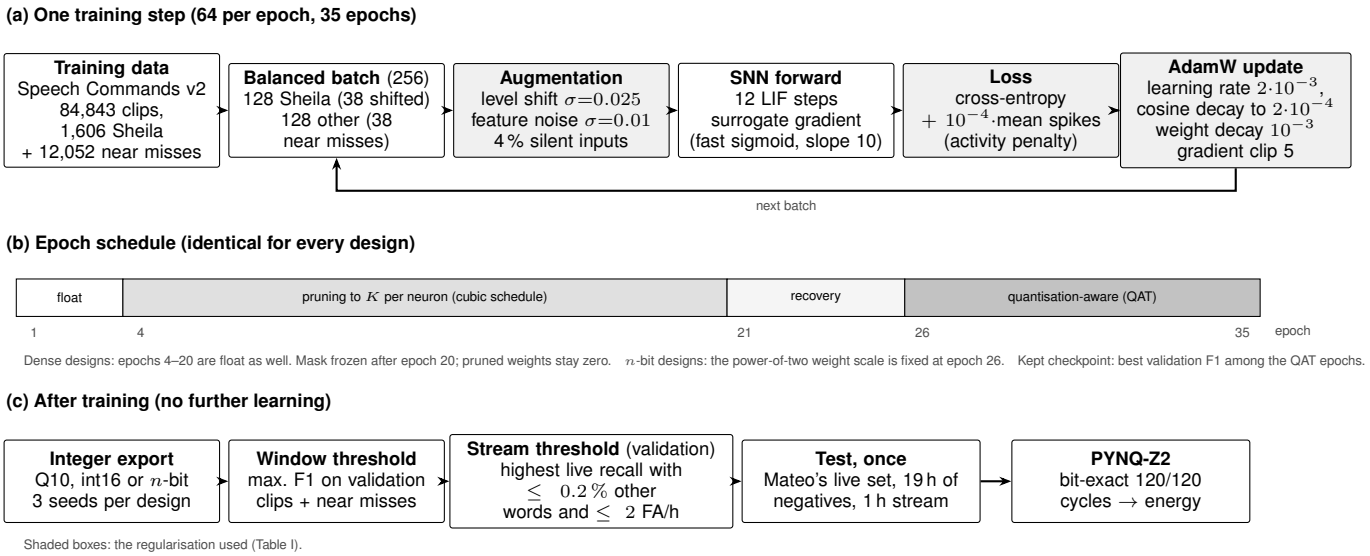


Fig. 3. How every design was trained and evaluated. (a) One update step; shaded boxes act as regularisers. (b) The 35-epoch schedule: pruned designs remove their smallest input synapses gradually in epochs 4–20, every design trains quantisation-aware in the last 10 epochs. (c) Thresholds are chosen on validation data only; the test sets are used once.

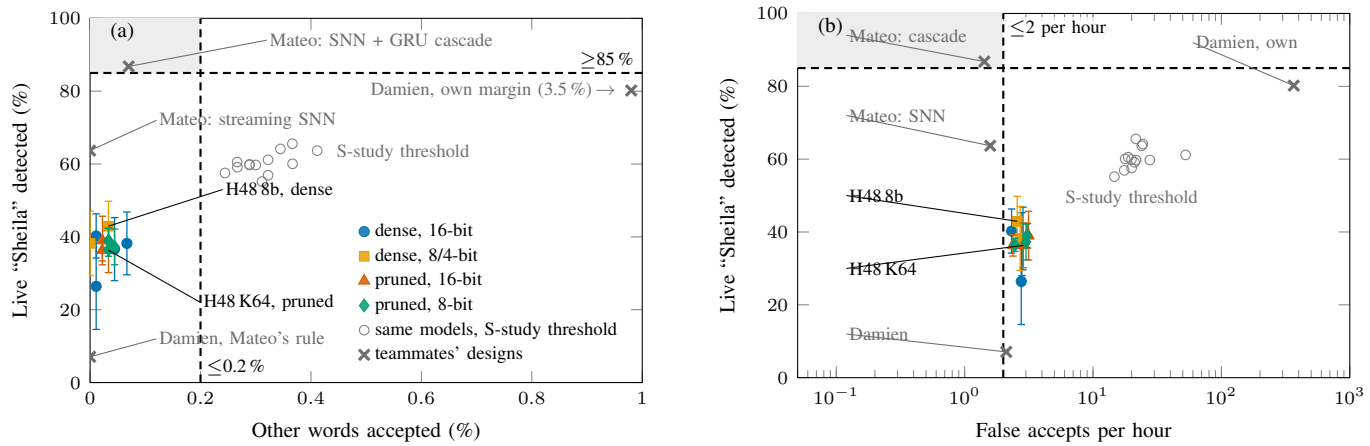


Fig. 4. Detection quality against the requirements; the shaded corner meets both. Filled markers: our designs at the threshold chosen on validation with Mateo's rule (mean ± sd over 3 seeds). Hollow markers: the same models at the earlier S-study threshold. Crosses: teammates' designs on the same test sets. Damien's model at its own margin lies at 3.5 % other words, off the scale in (a).

- The difference, 6.6 points, is within the dense design's spread (6.9 points), so the **pruned design is recommended**: 42 % less energy and 83 % less memory per inference. The choice is the same with the pooled spread and with test F1 as the measure. Its 8-bit version (37.3 %) is equivalent.
- Caveat: `ksdot` exists only as an experimental bitstream; the deployed model is the dense one.

VI. Trade-offs and Key Findings

- **Detection, not cost, is the limit.** Every design meets five of the seven requirements with margins of 2 to more than 500 times, and misses the two detection requirements.
- **Running speech sets the threshold.** Most false accepts are LibriSpeech sentences; keeping them at 2 per hour costs about 20 points of live detection (65.6 to 42.9 % for the deployed model).
- **Size and synapse levers buy cost, not accuracy.** Since none of them moves the detection, the cheapest design is the right one: pruning to 64 synapses per neuron with sparse hardware.
- **The power budget is not binding.** Static power and the block RAM set the PL power (93 mW floor with clock gating); the network adds at most a few milliwatts.
- **A one-window network is not enough.** Mateo's streaming SNN with a GRU verifier meets the detection requirements on the same tests (86.8 %, 1.42 per hour, 0.07 %) with 117 ms worst-case compute. Our designs mark the low-cost end: about 100 μJ and 12 KiB per window.

VII. Next Steps

- Train on Mateo's full data (MSWC near misses, Piper TTS, LibriSpeech negatives, rooms and microphones) to cut the running-speech false accepts.

TABLE III

ALL DESIGNS AT THE THRESHOLD CHOSEN ON VALIDATION (MEAN OF 3 SEEDS). TEST F1 IS THE CLIP-LEVEL F1 ON THE FULL SPEECH COMMANDS TEST SPLIT AND ON DAMIEN'S 3,212-CLIP SUBSET. THE COST COLUMNS GIVE THE MOST EFFICIENT HARDWARE MEASURED FOR EACH DESIGN. MET: NUMBER OF THE SEVEN REQUIREMENTS MET.

	Test F1		Live	Other	FA	Latency		$t_{\max}$	Memory	$E_{\inf}$	P_{PL}	
Design	full	subset	(%)	(%)	per h	(ms)	Hardware	(ms)	(KiB)	(μJ)	(mW)	Met
Requirement			$\geq$85	$\leq$0.2	$\leq$2	$\leq$500		$\leq$200	$\leq$144		$\leq$1000	
H16	0.764	0.813	26.4 ± 11.8	0.01	2.76	204	`kdot`+table	0.16	24.1	45	367	5/7
H32	0.760	0.821	36.6 ± 8.6	0.04	2.79	179	`kdot`	1.38	48.3	380	367	5/7
H48	0.790	0.832	40.3 ± 6.1	0.01	2.32	175	`kdot`+table	0.59	72.4	165	367	5/7
H95	0.781	0.830	38.2 ± 8.6	0.07	2.85	160	`kdot`+table	1.15	143.2	323	367	5/7
H48 8b	0.789	0.833	42.9 ± 6.9	0.03	2.57	169	`kdot`+table	0.59	72.4	165	367	5/7
H48 4b	0.785	0.846	38.2 ± 8.8	0.00	2.71	193	`kdot`+table	0.59	72.4	165	367	5/7
H48 K512	0.773	0.820	39.0 ± 6.7	0.02	3.15	175	`ksdot`+table	0.56	96.4	157	367	5/7
H48 K256	0.780	0.841	39.0 ± 3.4	0.03	3.09	170	`ksdot`+table	0.44	48.4	123	367	5/7
H48 K128	0.784	0.839	36.3 ± 2.9	0.02	2.37	176	`ksdot`+table	0.38	24.4	105	367	5/7
H48 K64	0.782	0.837	36.3 ± 6.1	0.03	2.86	134	`ksdot`+table	0.35	12.4	97	367	5/7
H48 8b K256	0.783	0.846	39.0 ± 3.3	0.03	3.06	171	`ksdot`+table	0.44	48.4	123	367	5/7
H48 8b K128	0.780	0.828	36.5 ± 1.8	0.03	2.46	162	`ksdot`+table	0.38	24.4	105	367	5/7
H48 8b K64	0.783	0.843	37.3 ± 4.9	0.04	3.01	141	`ksdot`+table	0.35	12.4	97	367	5/7

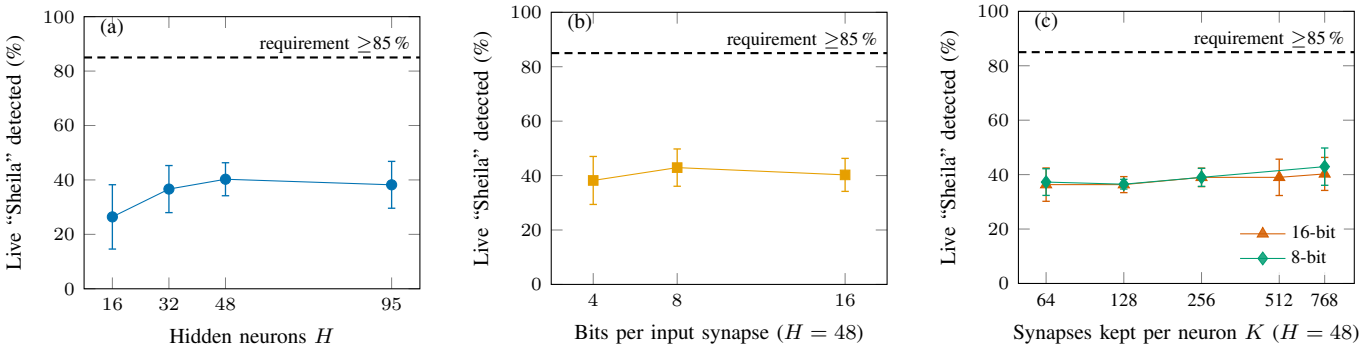


Fig. 5. Live detection at the requirement threshold against (a) the number of neurons, (b) the bits per input synapse and (c) the synapses kept per neuron ($K = 768$: dense). Mean $\pm$ sd over 3 seeds. Above 16 neurons no lever changes the detection by more than the seed spread, and none comes near 85 %.

- Add time awareness (Mateo's streaming SNN and GRU verifier) on top of the pruned front end.
- If the pruned design is adopted: build `ksdot` into the release bitstream.

Limits. Three seeds per design (live-detection spread 2–12 points); false accepts per hour come from 19 h of audio, so 2 per hour means about 38 events (± 6); energy is a Vivado estimate (validated within 7 % on a group design).

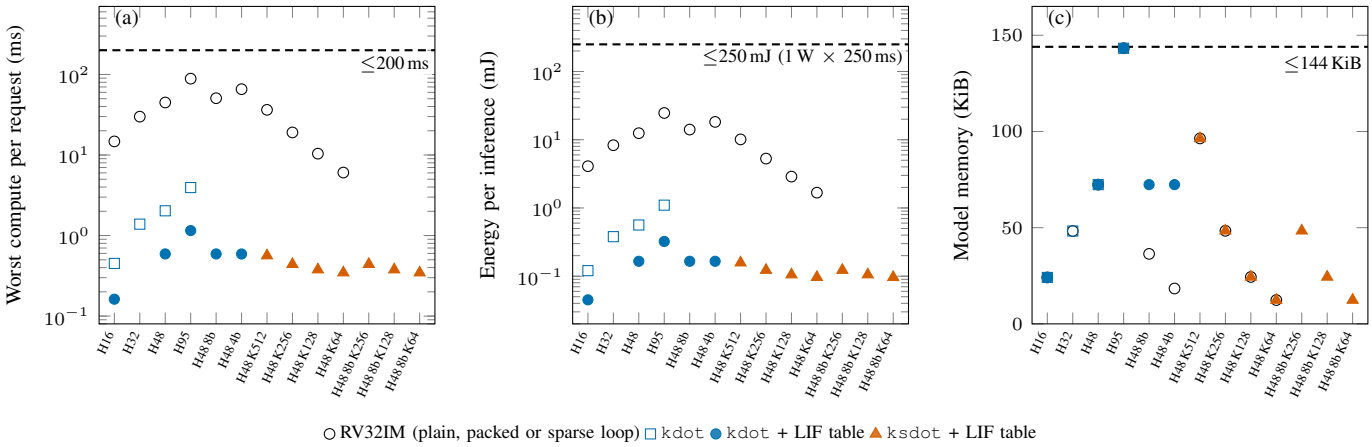


Fig. 6. Cost requirements for every design and hardware (mean of 3 seeds; cycles from bit-exact board runs, energy from Vivado power estimates). Dashed: the requirement. Every design meets all three; the energy budget is met by a factor of 10 or more.

TABLE IV
THE TWO CHOSEN DESIGNS AGAINST THE REQUIREMENTS, NEXT TO MATEO'S DESIGNS ON THE SAME TEST SETS (MATEO'S VALUES FROM HIS FINAL TEST REPORT; COMPUTE FROM HIS README; HIS MEMORY AND POWER ARE REPORTED IN HIS OWN SECTIONS). √ MET, × MISSED.

Requirement	Target	H48 8b `kdot`+table	H48 K64 `ksdot`+table (recommended)	Mateo streaming SNN	Mateo SNN + GRU
Live "Sheila" detected	≥85 %	42.9 % ×	36.3 % ×	63.7 % ×	86.8 % √
False accepts per hour	≤2	2.57 ×	2.86 ×	1.58 √	1.42 √
Other words accepted	≤0.2 %	0.03 % √	0.03 % √	0.00 % √	0.07 % √
Word end to decision	≤500 ms	169 ms √	134 ms √	44 ms √	175 ms √
Compute per request	≤200 ms	0.59 ms √	0.35 ms √	10.30 ms √	117.00 ms √
Model memory	≤144 KiB	72.4 KiB √	12.4 KiB √	—	—
Average PL power	≤1 W	367 mW √	367 mW √	—	—

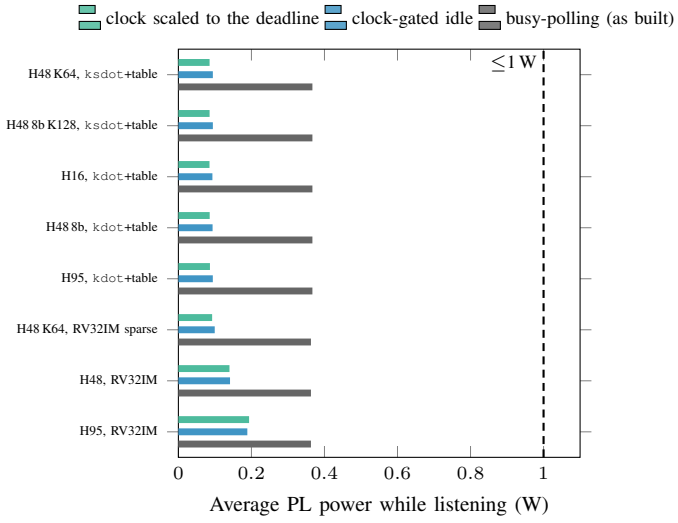


Fig. 7. Average power of the programmable logic at one window per 250 ms. Busy-polling is the firmware as built; clock gating and clock scaling are calculated. Every design is far below the 1 W budget: the static power and the block RAM, not the network, set the power.

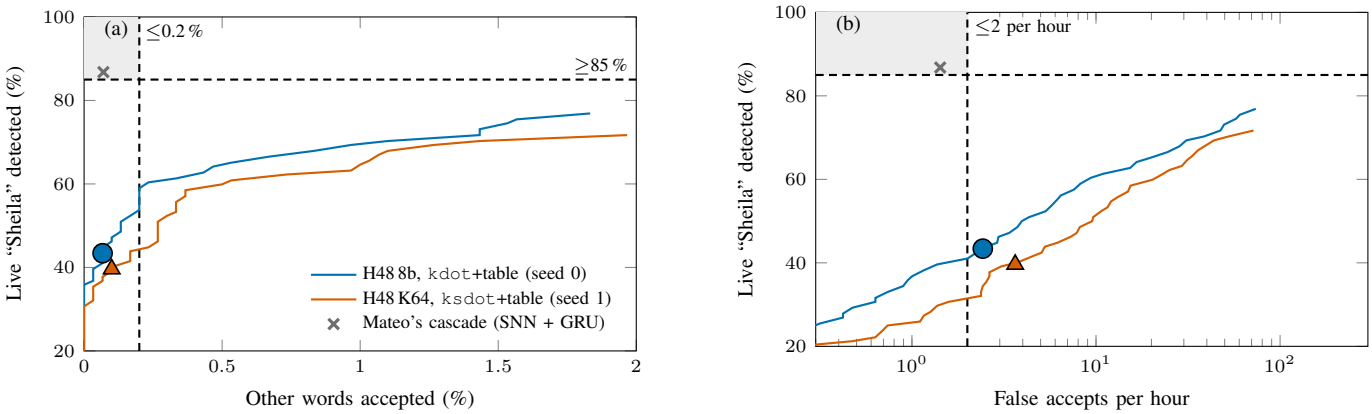


Fig. 8. What the two chosen designs can trade (one seed each, the one closest to the 3-seed mean). Lines: the test result at every threshold; large markers: the threshold fixed on validation. Even at 2 % other words (ten times the limit) and about 70 false accepts per hour, live detection stays below 80 %: these one-window networks cannot reach 85 % at any threshold. Mateo’s cascade meets both requirements.

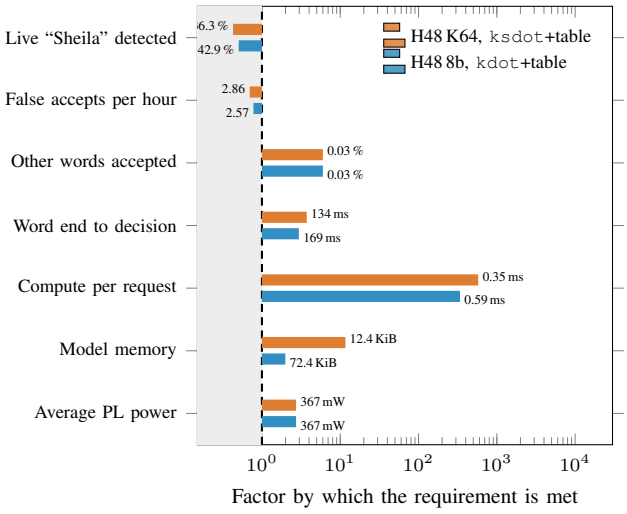


Fig. 9. The two chosen designs against every requirement, as the factor by which each is met (value/target for live detection, target/value otherwise); numbers are the measured values. Both meet the five cost and latency requirements by a wide margin and miss live detection and false accepts per hour.